



RESEARCH

& INNOVATION UPDATES

Smartwatch that measures stress hormone

A hormone called cortisol gets produced in the human body, which is key to identifying conditions such as depression and post-traumatic stress when levels of the hormone increase. Cortisol levels traditionally have been evaluated through blood samples. While those measurements help diagnose certain diseases, they fail to capture changes in cortisol levels over time. Now, a UCLA research team has developed a smartwatch that assesses cortisol levels found in sweat—accurately, noninvasively, and in real time. In the new smartwatch, a strip of thin adhesive film collects tiny volumes of sweat. An attached sensor detects cortisol using engineered strands of DNA, called aptamers. When cortisol attaches, the aptamer changes shape in a way that alters electric fields at the surface of a transistor.

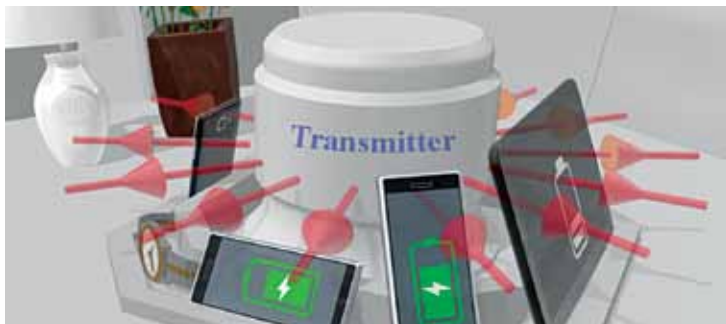
Cortisol is well-suited for measurement through wearable devices, because its concentration levels in sweat are similar to its circulating levels (Credit: <https://newsroom.ucla.edu>)



Wireless charging for home use

Existing wireless charging systems are known to transfer power in a specific direction or position. Now, researchers

at Aalto University have developed a new wireless charging system that can charge devices placed anywhere within a certain radius around it. For this, a cylindrical power coil has been designed that has wires at the top and bottom wound in the opposite directions, with a z-shaped bridge connecting them. Current flows through these windings in opposite directions, producing complementary magnetic fields. This results in an even magnetic field around the middle of the charging coil, allowing receivers placed anywhere within that area to charge efficiently, regardless of their position or orientation. By providing a donut-shaped charging field, the new system offers a more convenient and reliable design for consumer use.



A new wireless charging technology opens the door to developing charging stations for consumer use (Credit: www.aalto.fi/en)